Study-design glossary and work status

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The first journal milestone covers CSS code-capacity, correlated quaternary code-capacity, and circuit-level decoding in sequence. This glossary separates completed implementation and evidence from cancelled or inconclusive work and planned research.

Status key

EXECUTED	Implemented, run, or checked at the scope stated; a pilot is not automatically a paper reproduction.
ON HOLD	Cancelled, inconclusive, or awaiting a source qualification; retained partial evidence remains visible.
FUTURE	Designed or required, but not yet implemented or assessed at this scope.

Source and claim boundary

The editable Markdown contains direct repository links. This PDF is a dated rendering of those definitions; future phase plans are intentions, not completed experiments. Phase 4's cancelled comprehensive campaign is retained only as partial evidence.

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Study sequence at a glance

Workstream	Status	Design role and current boundary	Source
Phases 1–4: CSS foundation	**Executed**	Initial literature map, native CSS baseline, canonical binary BP, update/schedule policies, memory policies, and Relay mechanisms have passed their phase gates.	Roadmap
Phase 5: guided interventions	**Future**	Fourteen verified plans cover BPGD, impulse, freezing, syndrome/sign flips, restarting, and a bounded B1 mechanism pilot. No Phase 5 plan has executed.	Phase 5 plans
Cancelled Phase 4 comprehensive evaluation	**On hold**	The oversized fresh evaluation and factorial were stopped. Their retained prefix is a receipt without a qualified comparison denominator; a later CSS campaign needs a new design.	Cancellation
Phases 6–9: CSS search, fallback, campaigns, comparison	**Future**	Beam search, optional postprocessing, preregistered campaigns, and CSS reproduction/controlled comparisons remain staged work.	Roadmap

Workstream	Status	Design role and current boundary	Source
Phases 10–11: joint Pauli decoding	**Future**	A compatible four-state kernel and matched correlated-versus-separate comparisons remain to be built and assessed.	Roadmap
Phases 12–13: circuit decoding	**Future**	Detector/observable semantics, native/canonical circuit runs, and circuit-paper comparisons remain to be built and assessed.	Roadmap
Phase 14: completed map and manuscript	**Future**	The full systematic map, supported-experiment release set, and three-regime journal evidence remain to be completed.	Roadmap

Research frame and comparison design

Term	Status	Meaning in this study	Source
Systematic mapping protocol	**Executed**	Versioned questions, search/screening, extraction, provenance, and an initial source-linked corpus. The completed map and publication cutoff are still future work.	Requirements
Completed systematic map	**Future**	The final selected corpus, flow/counts, extraction dictionary, limitations, and updated search through a declared cutoff.	Phase 14
Experiment registry	**Executed**	A versioned manifest links a supported run to source identity, decoder settings, artifacts, samples, software, budgets, expected outputs, and discrepancies. The release-wide supported set is future work.	Project
Canonical kernel family	**Executed**	One inspectable binary CSS execution engine currently owns compatible BP policy behavior. A joint quaternary kernel and circuit integration require separate compatibility work.	Architecture
Policy composition	**Executed**	Explicit bindings for updates, schedule, memory, intervention, stopping, candidate selection, and postprocessing describe compatible decoder behavior. Some policy families remain future.	Canonical engine
Source-faithful configuration	**Future**	A paper-specific run must preserve its source-defined problem, algorithm, settings, implementation identity, and deviations before a reproduction claim. Source-qualified references already exist at narrower pilot scope.	Project
Controlled comparison	**Future**	A second analysis view holds samples, code, work limits, and evaluation rules common while varying declared decoder mechanisms. It answers a different question from paper-specific reproduction.	Project

Term	Status	Meaning in this study	Source
Matched physical samples	**Executed**	Identified error draws can be reused across eligible arms so differences are attributable to policy choices. Existing pilots use this design; the Phase 5 B1 matched pilot is only planned.	Phase 4 guide
Preregistration and stopping rule	**Future**	Freeze targets, tolerances, sample/work budgets, denominator rules, and stop criteria before publication campaigns; Phase 8 owns reusable campaign assessment.	Roadmap

Decoding problems and correctness

Term	Status	Meaning in this study	Source
CSS code-capacity decoding	**Executed**	Separate binary X/Z decoding on validated CSS check and logical matrices under identified data-error samples. Existing pilots are code-capacity adaptations.	Project
Correlated quaternary decoding	**Future**	Joint inference over I/X/Y/Z Pauli states with Y and X/Z correlations preserved in a mathematically compatible four-state kernel.	Phase 10
Circuit-level decoding	**Future**	Decode noisy extraction circuits using detector and logical-observable semantics, not a relabeled CSS data-error model.	Phase 12
Check matrix and syndrome	**Executed**	The binary parity-check matrix maps an error to observed parity violations; original syndromes are immutable inputs to outcome validation.	CSS requirements
Ordered logical basis	**Executed**	Logical operators and their pairing are part of a code artifact's identity, so a matrix with changed logical ordering is not treated as the same artifact.	CSS requirements
Original problem	**Executed**	The immutable observed decoding instance against which a selected recovery is judged, even if a policy solved a changed working instance.	Canonical engine
Working problem and candidate lift	**Future**	Phase 5 will transform priors, graph, or syndrome for an attempt and map a working-space candidate back to original coordinates with a checkable certificate.	Phase 5 context
Exact recovery	**Executed**	The selected recovery equals the sampled physical error; this is one independently evaluated outcome class.	CSS requirements
Stabilizer-equivalent recovery	**Executed**	A non-identical recovery can still have the same logical action as the sampled error; the evaluator distinguishes it from exact recovery.	CSS requirements
Syndrome-valid logical failure	**Executed**	A candidate matches the original syndrome but differs by a nontrivial logical operator; parity success alone does not imply decoding success.	CSS requirements

Term	Status	Meaning in this study	Source
Syndrome-invalid candidate	**Executed**	A candidate fails the original parity check and cannot be promoted to success by convergence on a modified working syndrome.	CSS requirements

BP policies and interventions

Term	Status	Meaning in this study	Source
Message-update rule	**Executed**	Sum-product, min-sum, normalized min-sum, and offset min-sum are explicit numerical policies with stated saturation, ties, and update semantics.	Phase 3
Schedule	**Executed**	Flooding, layered, serial, residual, randomized, and source-defined cluster-sequential orders specify when fresh messages become visible and what work is counted.	Phase 3
Memory or damping policy	**Executed**	Damping, mixing, momentum, iteration history, and disordered memory declare operands, order, lifetime, and reset behavior in the binary CSS engine.	Phase 4 guide
Relay-BP	**Executed**	Source-linked multi-leg BP transfers a finite posterior-derived prior between legs and accounts for each leg and selection. Its completed Phase 4 gate is a mechanism result, not circuit-curve reproduction.	Phase 4 verification
BPGD (guided decimation)	**Future**	Planned source-qualified rounds select a reliable undecimated variable and change its prior while continuing BP. Zero/tie/final-round choices remain named interpretations.	Phase 5 research
Impulse decoding	**Future**	Planned baseline-failure trigger launches bounded independent forced-bit attempts, records candidate origin/work, and selects after original-problem validation. Structural shortening is separately labeled adaptation.	Phase 5 research
Freezing	**Future**	A bounded policy fixes eligible variables for a declared lifetime and records release or retry, rather than treating a temporary prior change as permanent truth.	Phase 5 context
Syndrome flip	**Future**	A source-linked attempt changes a working syndrome, carries a correction certificate, then checks the lifted result against the original observation.	Phase 5 research
Sign flip	**Future**	A declared target, trigger, sign/magnitude convention, and state consequences distinguish a posterior perturbation from a changed channel prior or message.	Phase 5 research
Restarting	**Future**	Explicit root/branch reset, preserved state, branch budget, and replay semantics. B1's exact distance is unknown, so its planned bounded restart is a mechanism adaptation.	Phase 5 research

Term	Status	Meaning in this study	Source
Beam search and ensemble	**Future**	Phase 6 will expose branch identity, masked BP, ranking/pruning, list selection, voting, and complete cost of accepted and discarded paths.	Phase 6
Optional OSD/LSD/SI/OTF	**Future**	Phase 7 will attach source-qualified extra processing after declared BP nonconvergence while retaining a separately inspectable BP-only outcome.	Phase 7

Provenance, evidence, and study gates

Term	Status	Meaning in this study	Source
Native reference	**Executed**	A pinned author/library implementation can be run as an executable comparison with declared translation and unavailable internal counters. This status applies only to references actually pinned and run.	Project
Unverified BPGD/impulse native identity	**On hold**	Phase 5 research has not established an eligible exact author-code identity for these procedures; no unnamed package may be promoted to source-faithful reference.	Phase 5 research
Experiment manifest	**Executed**	A strict, versioned saved specification preflights decoder, code, sample, source, numerical, budget, and output identities before sampling. Phase 5 extends it only after execution.	Registry requirements
Event ledger	**Executed**	Per-shot and control-flow evidence records attempts, interventions, candidates, selection, termination, missingness, and typed work; optional detailed traces do not replace the ledger.	Ledger requirements
Replay and independent verification	**Executed**	A verifier reconstructs canonical terminal claims from persisted evidence while excluding nondeterministic timing from replay equality. New Phase 5 event semantics remain future.	Canonical engine
Candidate lineage	**Executed**	Stable IDs tie a selected original-space recovery to attempts, parents, and evidence without depending on append timing.	Canonical engine
Pilot adaptation / not_assessed	**Executed**	A runnable, source-linked mechanism pilot whose changed problem or settings are disclosed and whose outcomes carry no published-curve agreement verdict.	Phase 1 pilot
B1 Relay setting selection	**On hold**	The completed Phase 4 calibration retained all terminal shots, but B1 had no eligible setting under its frozen work cap; no selected B1 Relay performance result follows.	Phase 4 verification
Cancelled Phase 4 factorial and fresh evaluation	**On hold**	The stopped comprehensive campaign retains partial receipts and explicit missingness. Later CSS assessment must be newly scoped rather than resumed or inferred from the prefix.	Cancellation

Term	Status	Meaning in this study	Source
Phase 5 bounded B1 mechanism pilot	**Future**	Plans use one common baseline plus six single-intervention arms on matched samples, with separate calibration, sealed caps, visible failed/denied attempts, and no decoder ranking.	Phase 5 context
Publication-curve reproduction	**Future**	Statistical agreement with a paper's curve requires declared source settings, uncertainty, tolerance, denominator, and discrepancy assessment; bit-for-bit identity is not required.	Project
Journal evidence package	**Future**	The final manuscript and release must link the completed map, supported-paper manifests, open software, and qualified results from CSS, correlated quaternary, and circuit stages.	Phase 14

Outcome and resource vocabulary

Term	Status	Meaning in this study	Source
Logical error rate (LER)	**Executed**	A defined fraction of admitted shots with a logically wrong or unsuccessful selected recovery; interpretation depends on the stated denominator and problem regime. Existing bounded CSS pilot estimates are not paper-curve reproduction.	Phase 4 guide
Nonconvergence	**Executed**	BP fails to meet its declared stop rule within a bounded attempt. It remains a visible terminal outcome and may later trigger an intervention or optional fallback.	Canonical engine
Leaf work and total work	**Executed**	Actual BP, branch/relay, candidate, and fallback work is recorded at unique leaf scopes and aggregated once; a denied operation cannot be counted as completed work.	Ledger requirements
Runtime and resource context	**Executed**	Iterations, wall time, CPU/resource observations, and native missingness retain units and scope; unlike work vectors, time is not a deterministic replay equality fact.	Canonical engine
Inconclusive result	**Executed**	A retained calculation that cannot meet its prespecified selection or statistical criterion under a cap; it is neither silently discarded nor reported as a pass.	Phase 4 verification
Statistical campaign	**Future**	Phase 8 will prespecify tolerances, sample budgets, stopping, shard verification, uncertainty, and acceptance before population-level or publication claims.	Phase 8

Source index

The editable Markdown provides direct links. These are the unique repository files cited by the glossary.

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